Neural Heterogeneity of Word Response

Proposal?

Juee Naik

November 13, 2026

Gap

Studies of free recall describe the average response to a word: high gamma rises first in visual and later in prefrontal regions (word onset vs. baseline; Burke et al., 2014; Kucewicz et al., 2019), and alpha/beta and gamma power change at largely separate sites (Marks et al., 2021). Single responses vary widely around this average. What sets the size of a given response (which word it is, where it falls in the list, the inter-stimulus interval (ISI) before it, the word before it, where the electrode sits, or the state of the brain at that moment) has not been measured. Each factor has its own literature: visual responses shrink after short intervals and recover with longer ones (images after a blank, ECoG; Groen et al., 2022), activity before a stimulus shapes the evoked response (fMRI; He, 2013), activity drifts with a list's temporal context (recall vs. study patterns; Manning et al., 2011), and a brain-wide encoding state predicts recall (recalled vs. not recalled words; Ezzyat et al., 2017). Here all of them are measured on the same single responses, in 361 subjects, 1007 sessions and ∼200 words per session, with task structure modelled as in DeHaan and Halpern (TBD).

Aims

Aim 1. What sets the size and timing of the response to a word? On average, occipital high gamma rises at word onset ($d = 0.34$) and alpha falls (-0.10), with smaller changes elsewhere (Fig. 1A). Split the response to each word event (word on vs. word off, Cohen's d effect size) into subject, subregion, electrode, session, task, word and word-event components per region, with a random-intercept model. *Preliminary (up to 5 electrodes per subject and ROI; Fig. 1B):* most variance is unique to an electrode during a word event (alpha 71–86%, high gamma 56–89%). Which word it is explains almost none (<1%), and task structure little (≤3%). The largest structured part is a word event's deviation shared by all of an ROI's electrodes (alpha 11–25%, high gamma 6–15%). Fitted within single subjects (Fig. 1C), the split looks like the group's: the residual is 77% in alpha and 64% in high gamma, and what the group model assigns to subjects goes to electrodes and subregions.

For the timing of the response, 38–70% of the variance of latency is unique to an electrode during a word event and 3–11% is shared by an ROI, while 18–28% is stable to the subject, subregion and electrode (Fig. 2A). Word events whose occipital response is 111 ms later have a temporal response 40 ms later, limbic 26, parietal 22 and frontal 17 ms later, and no later hippocampal response (Fig. 2B, D): only 15–36% of a word's delay in visual cortex is passed on, so the occipital-to-frontal order of the average response is not a sequence that each word follows. Within occipital cortex,

the fusiform gyrus responds later and less strongly than lateral occipital cortex and lingual gyrus (Fig. 2C). Size and timing are separate too: the peak of a word event's high-gamma response barely relates to its latency ($r = 0.01$–0.04, ≤ 0.13 after correcting for noise; Fig. 2E).

Stable differences between sites are large only in occipital high gamma (subject 8%, electrode 22%; subregion about 3%), and there they replicate between session halves (Fig. S1A; median $r = 0.90$, 0.20–0.59 elsewhere). Outside occipital regions 68–76% of electrodes show neither an alpha nor a high-gamma response, against 35% in occipital regions; across electrodes the bands share the size of the response more than its direction (Fig. S1B; r of $|d|$ 0.46 in occipital regions, 0.13–0.21 elsewhere).

Planned: the same split for power during the ISI alone.

Aim 2. Is the shared momentary component a state that persists, and is it one state? During the ISI, occipital high-gamma power is higher when the ISI is short (Fig. 3A, B). Aligned to the previous word's offset rather than the next word's onset, the curves of all ISI lengths fall on one decay that ends about 300 ms after the offset (Fig. S2): the higher level is the tail of the response to the previous word disappearing, over well before the next word appears. After a long ISI the occipital response peak is larger ($+0.058$, $q < .001$) but not earlier (-4 ± 3 ms, n.s.; Fig. 3C), as recovery from adaptation would make it (Groen et al., 2022). At the same electrode, with serial-position means removed, the previous word's response correlates with the current one in every ROI (Fig. 3D; high gamma $r = 0.06$–0.12, alpha 0.03–0.05), more than the response two words back does (by 0.014–0.024 and 0.007–0.020): a slow drift plus a carry-over from the previous word, which does not change with the length of the ISI (long − short $|\Delta r| \leq 0.008$, n.s.). The two bands' shared word-event components are each reliable (split-half r 0.17–0.39) yet barely related (-0.13 to 0.19 after correcting for noise): two momentary states, not one.

Planned: whether power during the ISI before each word event predicts its response; whether the shared word-event component is shared across regions.

Aim 3. Does the momentary state matter for memory? Planned: whether the shared word-event component and ISI power, rather than the stable site differences or task structure, predict whether a word is recalled (Fig. 4).

Aim 4. Does connectivity between regions vary in the same way? At word onset alpha ciPLV rises in 9 of 12 regions, with hubs first in occipital regions (0–400 ms) and then in limbic, parietal, frontal and temporal regions (Fig. 5A).

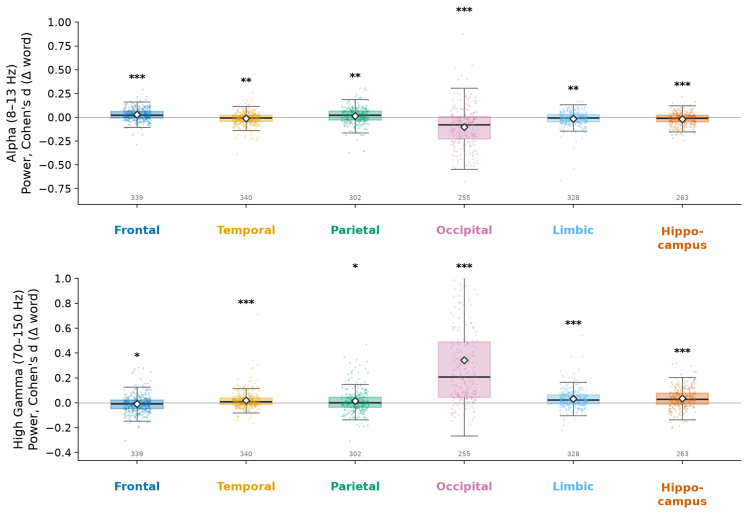
Planned: the variance split of Aim 1 for synchrony (Fig. 5B).

Summary. The response to a word depends little on which word it is or where it falls in the task, and differences between locations are large only in occipital high gamma. The largest structured part is a momentary component shared across a region's electrodes, which persists over a few words. Word by word, timing beyond visual cortex is set largely anew rather than inherited from it.

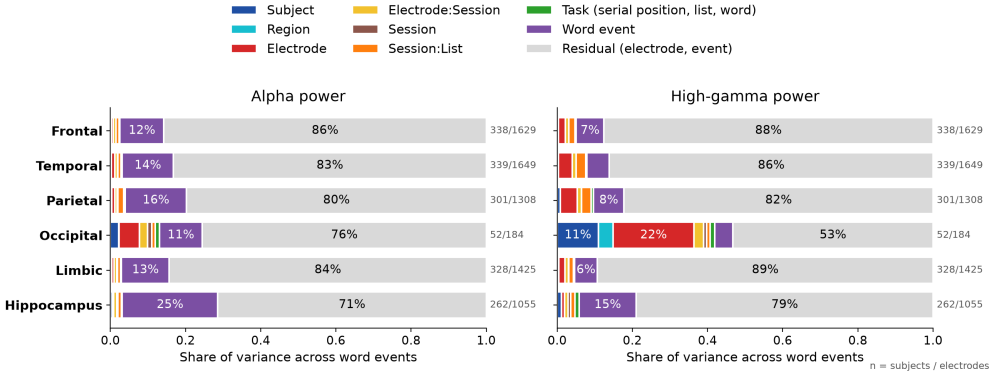
Figures

Supplement

A



B (Prelim: all subjects, ≤5 random electrodes per subject × ROI; occipital row: 52 random subjects, with subregion)



C (Prelim: occipital, the 99 subjects with ≥5 electrodes and ≥2 sessions, all their electrodes)

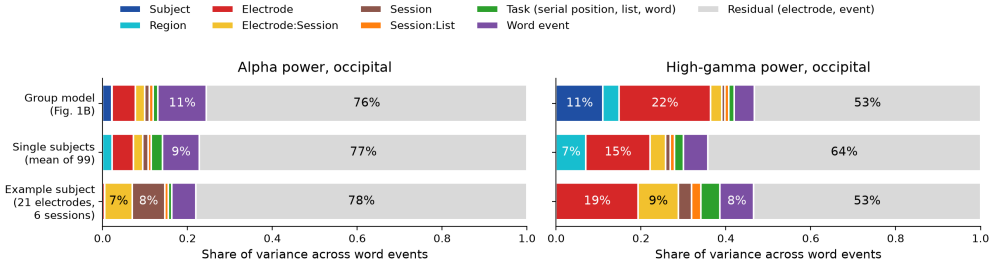


Figure 1: (A) Power change at word onset (Cohen's d, word on vs. off, across subjects). (B) Variance components of the response to each word event, one mixed model per ROI; occipital: with anatomical subregion as a component (52 subjects). (C) Occipital shares from the group model (B's occipital row), averaged over single-subject fits (the same model without subject terms) and for the subject with the most electrodes × sessions.

A (Prelim: all subjects, ≤5 random responsive electrodes per subject × ROI, single-event cross-correlation latency; frontal, temporal, parietal, occipital rows: 24–52 random subjects, with subregion)

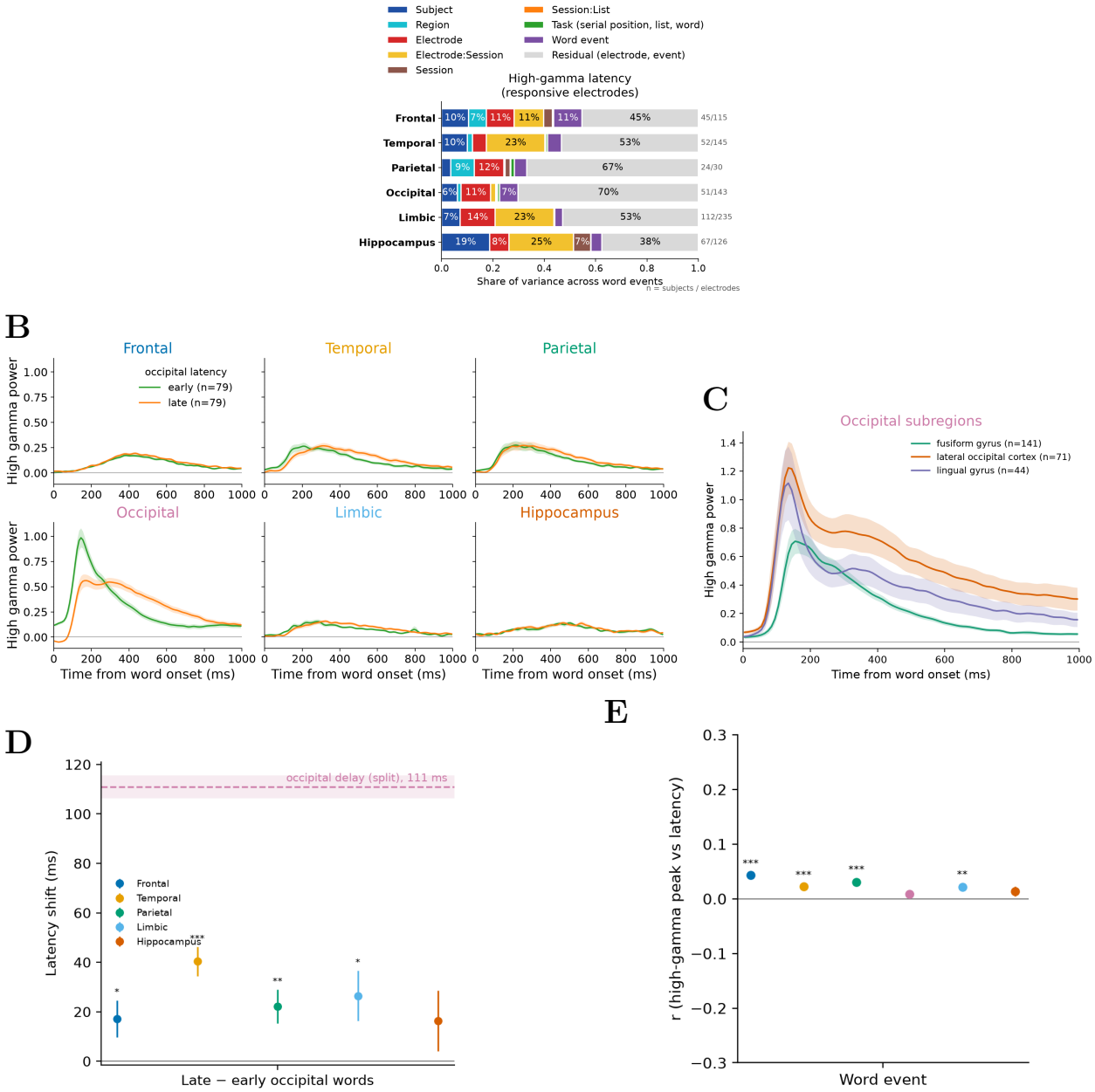


Figure 2: (A) Variance components of high-gamma latency per word event (cross-correlation with the electrode's average response, responsive electrodes); frontal, temporal, parietal and occipital: with anatomical subregion (regions with ≥30 subjects). (B) High gamma per region for word events whose occipital latency (cross-correlation with each electrode's average response) is early vs. late (session median split); mean ± SEM across subjects. (C) Occipital high gamma per anatomical subregion (the three with the most subjects, responsive electrodes); legend: subjects. (D) Lag of each region's late-word curve behind its early-word curve (cross-correlation per electrode); dashed: occipital itself, the delay set by the split (± SEM); subjects with responsive electrodes in both occipital regions and the region (41–112). (E) High-gamma peak vs. latency (cross-correlation) across word events at the same electrode (serial-position and list means removed). D, E: mean across subjects ± SEM per region, stars: FDR.

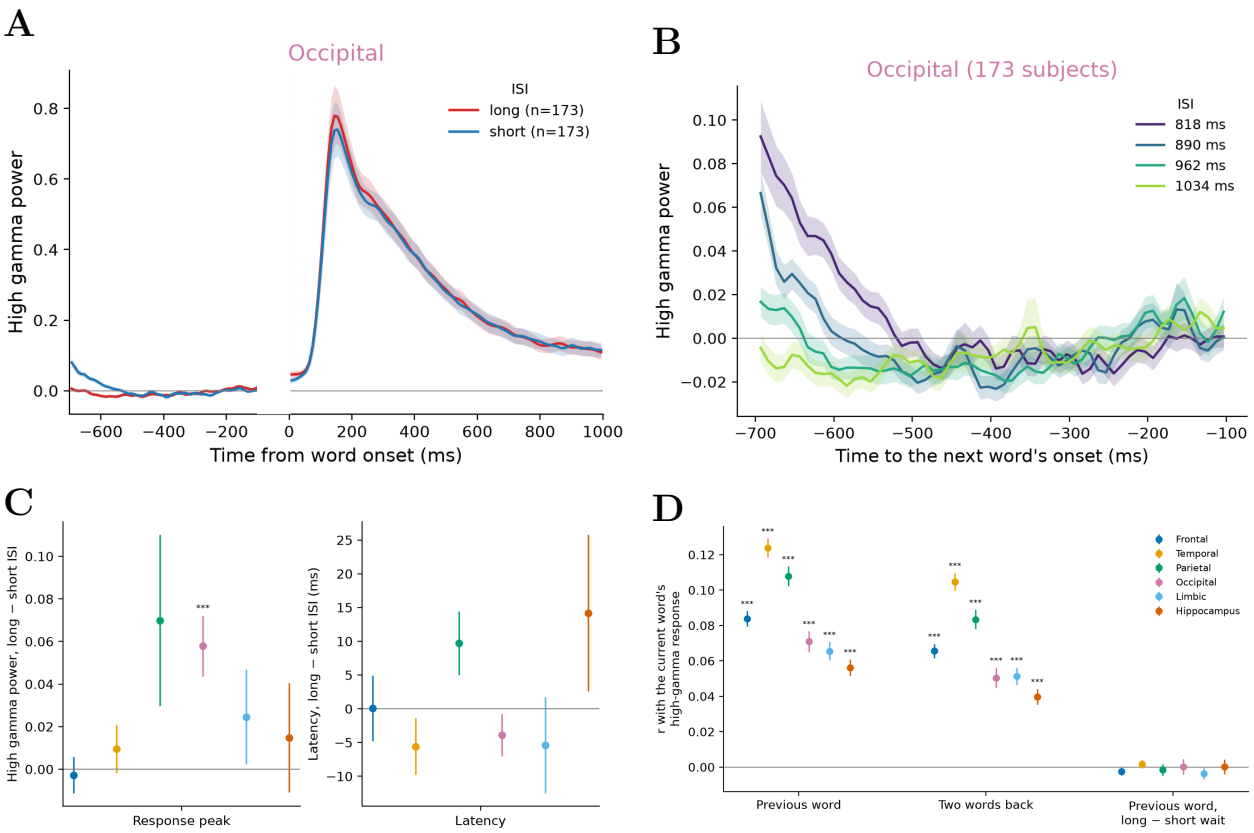


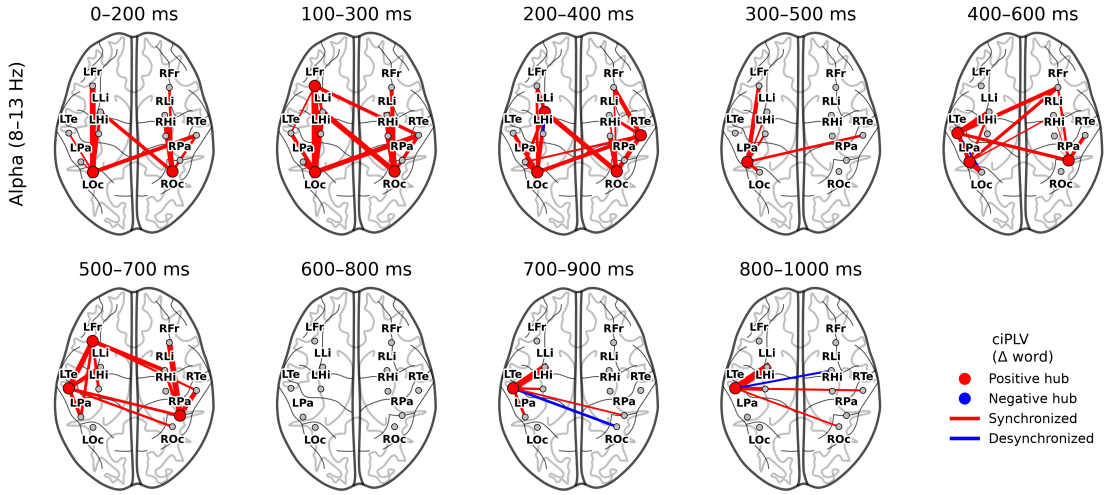
Figure 3: (A) Occipital high gamma after long vs. short ISIs (session median split), from the end of the ISI (−700 to −100 ms) through the response (responsive electrodes). (B) Occipital high gamma during the ISI per ISI quartile, aligned to the next word's onset. (C) Long − short response peak and latency (lag by cross-correlation). (D) High-gamma response of each word vs. the word one and two back in the same list (same electrode, serial-position means removed), and the change after long vs. short ISIs. Mean across subjects ± SEM; stars: FDR.

(Planned)



Figure 4: Recall predicted by the word-event component and ISI power vs. the stable site differences and task structure.

A



B (Planned)

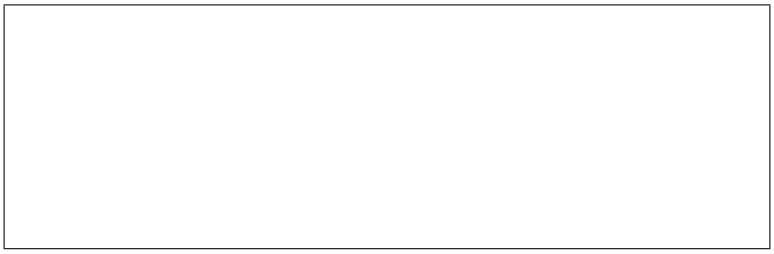
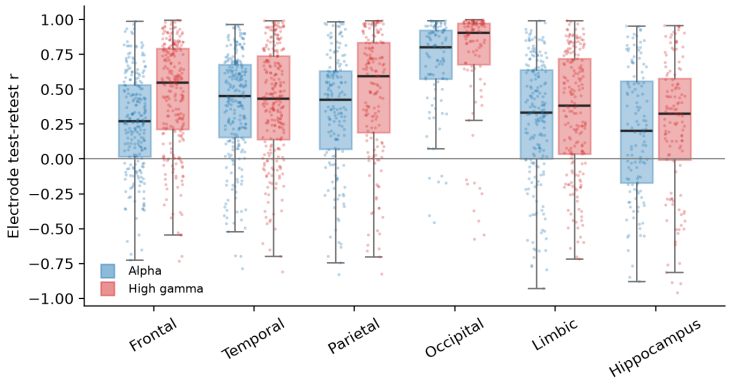


Figure 5: (A) Alpha ciPLV change at word onset in 200 ms windows: hubs (nodes) and each hub's five strongest connections. (B) Variance components of synchrony.

A



B

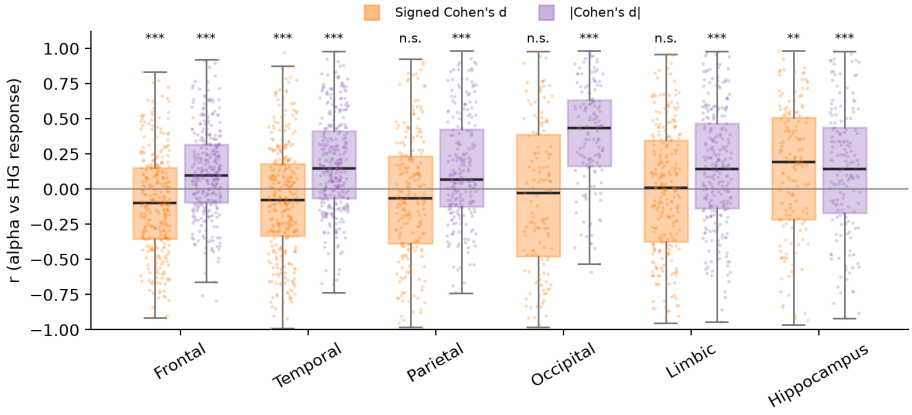


Figure S1: (A) Electrode test-retest: r of electrodes' Cohen's d between session halves (1 dot = 1 subject). (B) Alpha vs. high gamma across an ROI's electrodes: r of Cohen's d and of $|d|$ per subject (dots); stars: subjects' mean r vs. 0, FDR.

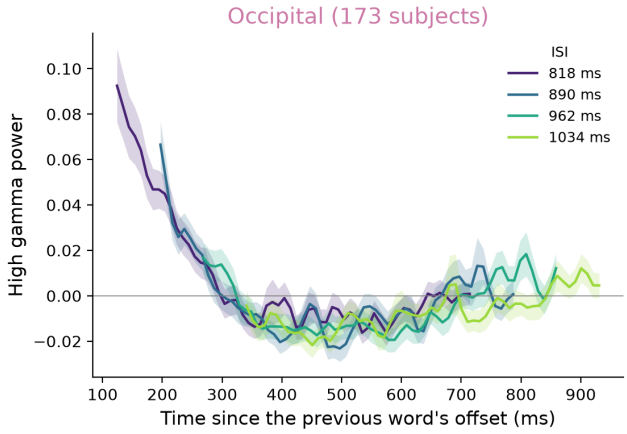


Figure S2: Fig. 3B's curves, each shifted by its quartile's mean ISI: time since the previous word's offset.

References

Burke, J. F., Long, N. M., Zaghloul, K. A., Sharan, A. D., Sperling, M. R., and Kahana, M. J. (2014). Human intracranial high-frequency activity maps episodic memory formation in space and time. *NeuroImage*, 85:834–843.

DeHaan, R. D. and Halpern, D. (TBD). *On the specificity, causality, and reliability of electrophysiological correlates of human memory: limitations and opportunities*. PhD thesis, University of Pennsylvania.

Ezzyat, Y., Kragel, J. E., Burke, J. F., Levy, D. F., Lyalenko, A., Wanda, P., O'Sullivan, L., Hurley, K. B., Busygin, S., Pedisich, I., Sperling, M. R., Worrell, G. A., Kucewicz, M. T., Davis, K. A., Lucas, T. H., Inman, C. S., Lega, B. C., Jobst, B. C., Sheth, S. A., Zaghloul, K., Jutras, M. J., Stein, J. M., Das, S. R., Gorniak, R., Rizzuto, D. S., and Kahana, M. J. (2017). Direct brain stimulation modulates encoding states and memory performance in humans. *Current Biology*, 27(9):1251–1258.

Groen, I. I. A., Piantoni, G., Montenegro, S., Flinker, A., Devore, S., Devinsky, O., Doyle, W., Dugan, P., Friedman, D., Ramsey, N. F., Petridou, N., and Winawer, J. (2022). Temporal dynamics of neural responses in human visual cortex. *The Journal of Neuroscience*, 42(40):7562–7580.

He, B. J. (2013). Spontaneous and task-evoked brain activity negatively interact. *Journal of Neuroscience*, 33(11):4672–4682.

Kucewicz, M. T., Saboo, K., Berry, B. M., Kremen, V., Miller, L. R., Khadjevand, F., Inman, C. S., Wanda, P., Sperling, M. R., Gorniak, R., Davis, K. A., Jobst, B. C., Lega, B., Sheth, S. A., Rizzuto, D. S., Iyer, R. K., Kahana, M. J., and Worrell, G. A. (2019). Human verbal memory encoding is hierarchically distributed in a continuous processing stream. *eneuro*, 6(1):ENEURO.0214–18.2018.

Manning, J. R., Polyn, S. M., Baltuch, G. H., Litt, B., and Kahana, M. J. (2011). Oscillatory patterns in temporal lobe reveal context reinstatement during memory search. *Proceedings of the National Academy of Sciences*, 108(31):12893–12897.

Marks, V. S., Saboo, K. V., Topçu, Ç., Lech, M., Thayib, T. P., Nejedly, P., Kremen, V., Worrell, G. A., and Kucewicz, M. T. (2021). Independent dynamics of low, intermediate, and high frequency spectral intracranial eeg activities during human memory formation. *NeuroImage*, 245:118637.